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# Recent Advances in Analog Front-End Architectures for ECG Signal Acquisition

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**ABSTRACT** This review provides a comprehensive and structured analysis of electronic interface systems for electrocardiogram (ECG) signal acquisition, with a particular emphasis on the design of analog front-end circuits. These front-end systems represent the first and most critical stage in any ECG acquisition chain, as they must handle biopotential signals that are extremely low in amplitude, bandwidth-limited, and highly susceptible to environmental interference. After the description of general and innovative principles of ECG acquisition, some examples of sensor electrodes, as well as proposed solutions of ECG signal acquisition are given. Starting from electrode technologies, including wet, dry, and textile-based solutions, their impact on interface design is discussed in terms of impedance, motion artifacts, and patient comfort. The fundamental building blocks of ECG front-ends—instrumentation amplifiers, driven-right-leg (DRL) circuits, and filtering stages—are then investigated, highlighting design strategies and innovative proposals. In particular, the fundamental design requirements for amplifiers have been examined: the discussion will be centered on traditional voltage-mode architectures as well as emerging current-mode approaches, analyzing their advantages and limitations in terms of power efficiency, gain stability, and integration capability, as presented in recent literature for ECG interface design. Future perspectives are outlined, covering the role of advanced CMOS technologies in the development and design of analog systems for biosignal monitoring, as well as the needs of electrodes based on nanostructured and biocompatible maerials. Through an in-depth examination of recent literature, we have compared some state-of-the-art solutions and outlined future trends for the development of high-performance, low-power ECG analog acquisition systems, particularly in the context of portable and wearable medical devices.

**INDEX TERMS** Analog biosignal processing, current-mode (CM) approach, driven-right-leg (DRL), electrocardiogram (ECG), ECG monitoring, sensor electrode, voltage-mode (VM) approach.

## I. INTRODUCTION

The electrocardiogram (ECG) is one of the most established, widespread, and reliable diagnostic techniques for analyzing the electrical activity of the heart. By recording the electrical potential generated by myocardial activity, the ECG can detect changes in heart rhythm and morphology, providing valuable information for the diagnosis and monitoring of a wide range of pathological conditions, including arrhythmia, ischemia, atrioventricular blocks, and electrolyte disturbances.

In recent years, the growing popularity of wearable devices and remote monitoring technologies has driven the evolution of the ECG from a traditional diagnostic system, typically confined to hospital settings, to a tool that is accessible in home, sports, and remote care contexts. This transformation requires suitable sensor electrodes and new architectural designs of acquisition systems, with particular attention paid to miniaturization, reduced noise, and power consumption and to the robustness of the signal acquired in real and potentially noisy conditions.

The heart of each ECG system is the electronic interface, which is the circuit block responsible for detecting, conditioning, and subsequently, if necessary, digitizing the biopotential signal. The role of the interface is fundamental to ensuring the reliability of the measurement: the ECG signal has a typical amplitude between 0.5 and 5 mV and requires typical needs to avoid attenuation and distortion; it is extremely vulnerable to external noise, such as common mode noise, electromagnetic interference from the mains (50/60 Hz), and motion artifacts.

The challenges associated with biopotential signal acquisition have driven research toward the adoption of increasingly difficult but efficient circuit solutions. Traditional voltage-mode (VM) systems, based on the use of operational amplifiers and linear architectures, represent the historical standard and are still widely used today in commercial and clinical applications. However, the intrinsic limitations related to bandwidth, stability, and scalability have led to the exploration of alternative approaches. In particular, the use of current-mode (CM) circuits, which employ active devices such as second-generation current conveyors (CCII) or second-generation voltage conveyors (VCII), has shown good perspectives for the implementation of high-performance front-ends capable of offering extended bandwidth, low noise, and low power consumption.

In this context, this review article aims to critically analyze recent scientific literature on electronic interface systems for ECG, with particular attention to the circuit approaches and to the analog active blocks used for signal conditioning. After an overview of ECG signal requirements and the desired characteristics of a biomedical front-end, some VM and CM architectures are examined, considering the associated advantages and critical issues. A classification of the most relevant electronic interfaces and a comparison of the results obtained in the most recent solutions, are also discussed.

## II. GENERAL PRINCIPLES OF ECG ACQUISITION

An ECG is a biopotential signal generated by the electrical activity of the heart during the cardiac cycle. The potential measured on the surface of the body, using electrodes positioned at strategic points, reflects the propagation of the myocardial depolarization and repolarization wave [1]. From an electronic point of view, this signal is characterized by a very small amplitude, typically between 0.5 and 5 mV, and a frequency between approximately 0.05 Hz and 100 Hz. Most of the diagnostic information is actually concentrated between 0.5 and 40 Hz, although in the presence of particular pathologies or physiological conditions, it may be useful to extend the useful range up to 150 Hz [2], [3].

The ECG trace has a well-known structure and is composed by P waves, QRS complexes, and T waves, each associated with specific electrical events in the heart [4], [5]. Acquiring this type of signal is not without its difficulties. The low-signal level, combined with the massive presence of environmental disturbances, requires careful design of the electronic interface. Among the main sources of noise that interfere with the ECG are electromagnetic interferences, particularly those related to the 50 or 60 Hz electrical network [6], [7], which manifest themselves in the form of common mode noise. In addition, patient movement or microsliding of the sensor electrodes can generate artifacts with amplitudes even greater than that of the useful signal [8], [9]. Added to this, there are low-frequency components due to thermal drifts or electrochemical polarization of the electrodes, which can introduce a dc offset of the order of hundreds of mV [10], [11].

One of the main challenges in measuring ECG signals is due to their low amplitude, making them highly susceptible to various types of noise and interference. In addition, the half-cell potential difference between the skin-electrode interface can reach 300 mV or even 500 mV, acting as a dc component that can saturate the amplifier input stage signals and limit its gain [1], [12], [13]. Muscle movement and motion artifacts due to suboptimal electrode contact with the skin can also generate significant interference signals, causing unstable offsets [14], [15], [16], [17]. ECG signals are also vulnerable to common-mode interference, which can be coupled to the human body and reach amplitudes of up to 1 mV.

To overcome these drawbacks, front-end amplifiers for ECG must face stringent requirements. High and stable gain is essential to amplify weak signals to raise the microvolt-level signal into ADC input range without distortion [18], [19]. At the same time, high common-mode rejection ratio (CMRR), typically greater than 60 dB, is required to suppress interference, particularly common-mode ones [12], [13], [20]. Low input-referred noise (IRN) is necessary for high-resolution measurements, especially at low frequencies where flicker noise is predominant [21], [22] so, the design must also ensure low power consumption to extend battery life in portable devices [23], [24], [25], [26].

To address these challenges, several design techniques have been developed, as follows.

1) AC coupling uses series blocking capacitors to remove the dc offset potential of the electrodes and ensure a correct dc level [27], [28], [29], [30].
2) Capacitively coupled instrumentation amplifiers (CCIAs) are widely used for their inherent ability to filter dc components and reduce low-frequency noise while improving energy efficiency [31], [32], [33], [34].
3) Chopper stabilization technique is used to drastically reduce 1/f noise and voltage/drift offsets [35], [36], [37]. Considering that flicker noise is particularly important in MOS transistors for low-frequency applications.
4) For ultralow power consumption, many designs exploit the subthreshold operation of transistors. This allows high transconductance ($g_m$) values to be achieved for a given bias current, increasing gain efficiency and reducing power consumption [38], [39], [40].
5) The dynamic threshold MOS technique has been proposed for ultralow voltage and power operational amplifiers, ensuring high gain [41], [42].

To improve CMRR and safety, ECG systems often employ a driven right leg (DRL) circuit [43], [44], [45], [46]. The DRL technique detects the common mode signal on the patient body and provides negative feedback to compensate for the body potential, so effectively reducing the common mode voltage to $\mu$V levels and counteracting electromagnetic interference. Fully differential amplifier topologies further enhance CMRR, provided their ground and supply rails are symmetrically laid out without unmatched components [47], [48].

High input impedance is a key requirement for reducing the load effect of the electrodes and rejecting the potential common mode signal resulting from impedance mismatches. Dry and capacitive electrodes are increasingly used for patient comfort and long-term monitoring, but they require amplifiers with extremely high input impedance [49], [50], [51].

Circuit techniques such as bootstrapping and active parasitic capacitance compensation can be employed to increase input impedance and minimize noise [52], [53], [54], [55], [56].

For dc suppression and resistance modeling, pseudoMOS resistors with a very high resistance value are often used, which is essential for implementing high-pass filters with very low cutoff frequencies, necessary for ECG signals [57], [58], [59], [60]. These elements make it possible to avoid the use of large on-chip resistors, which would occupy a larger area.

In addition to VM techniques, CM approach offers promising solutions to many of the challenges in designing amplifiers for biomedical signals [61], [62], [63], [64], [65]. This approach processes signals directly in the current domain, which can lead to numerous advantages. Among these, the most notable are high CMRR and wide bandwidth without the need for precisely matched resistors. This is particularly advantageous for reducing manufacturing costs and complexity. In addition, CM solutions tend to consume less power, and, in some cases, the bandwidth can remain constant and independent of gain.

Maximizing noise and power efficiency is another key objective. In this sense, employed approaches include the current reuse technique, which allows high transconductance to be achieved while reducing current consumption. Multistage transconductance operational amplifier (OTA) architectures are being explored to improve dc gain and figure of merit (FOM) [66], [67], [68], [69], [70].

In summary, ECG amplifier design focuses on integrated solutions with low power consumption, low noise, high gain, high CMRR, and high input impedance, often implemented with advanced CMOS techniques. These advances are necessary for the realization of portable and wearable medical devices that can effectively monitor bioelectrical signals in real-life scenarios.

## III. ELECTRONIC FRONT-END FOR ECG MONITORING

A typical ECG acquisition front-end is composed of several key functional blocks designed to ensure signal integrity, high CMRR, and adaptability to various clinical or wearable environments. The structure is generally composed of electrodes, instrumentation amplifier (IA), DRL circuit, filtering stages and, optionally, further programmable gain amplification as shown in the diagram of Fig. 1.

### *A. ECG SENSORS: ELECTRODES*

The acquisition process begins at the sensor electrode level, where biopotential signals originating from cardiac activity are collected from the skin surface. Traditional wet electrodes, usually made of silver/silver chloride (Ag/AgCl) combined

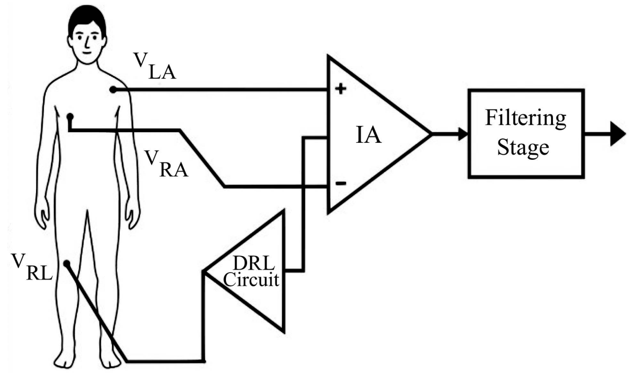


**FIGURE 1.** **ECG analog acquisition system diagram.**

with an electrolyte gel, are still considered the clinical gold standard [71]. The electrolyte reduces the skin–electrode impedance and ensures stable signal acquisition in short-term monitoring. They are widely used in hospitals due to their high fidelity; however, they suffer from limitations in continued applications, such as gel drying, skin irritation, and discomfort during long-term wear.

To overcome these limitations, dry electrodes have been developed. These electrodes do not require conductive gel and can be fabricated using metals (e.g., gold, stainless steel), conductive polymers (e.g., PEDOT:PSS), or composites such as carbon-based rubber [72]. Dry electrodes provide improved comfort and are more suitable for long-term monitoring and wearable systems. Nonetheless, they typically present higher skin–electrode impedance and increased susceptibility for motion artifacts, which demands careful analog front-end (AFE) design to maintain a suitable signal-to-noise ratio (SNR).

More recently, textile-based fabric electrodes have emerged as promising solutions for wearable health monitoring [73]. These electrodes are integrated directly into garments, typically using conductive yarns woven with conventional textile fibers. Their performance depends on both the fabric structure and the applied pressure on the skin. Wang et al. [74] systematically investigated fabric electrodes woven in plain, twill, and satin structures using silver-plated nylon yarns. Their study showed that the fabric structure significantly affects contact impedance and ECG quality. Plain weave electrodes, characterized by a compact structure, provided low impedance and stable ECG signals at rest. Twill weave electrodes, with higher elasticity and adaptability to body contours, demonstrated better performance during motion, with reduced artifacts. Satin weave electrodes, with smoother surfaces and fewer interlacing points, offered a good compromise between comfort and signal stability, particularly under dynamic conditions.

Contact pressure is another critical parameter in fabric electrodes. Too low pressures ($<2$ kPa) result in insufficient contact and high impedance, while excessive pressures ($>5$ kPa) may cause discomfort and degrade signal quality. The optimal range, identified between 3 and 4 kPa, ensures

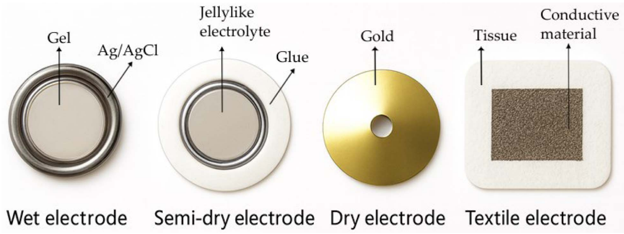


**FIGURE 2.** Common types of electrodes used for ECG acquisition.

both signal stability and wearer comfort [75]. Under static conditions, all three textile structures could acquire high-quality ECG signals comparable to gel electrodes. However, under dynamic movements such as torso expansion or arm lifting, the differences became more evident. At low pressures, plain weave electrodes often failed to capture recognizable QRS complexes, whereas twill and satin electrodes provided clearer and more stable signals, with satin weave demonstrating the best overall performance. Fig. 2 illustrates the most common types of electrodes used in ECG acquisition: wet Ag/AgCl electrodes, semidry electrodes with jellylike electrolyte, dry metal electrodes, and textile electrodes integrated into fabric substrates.

Electrode performance directly influences AFE design. Higher contact impedances require amplifiers with ultrahigh input impedance—while susceptibility to motion artifacts necessitates robust filtering strategies to suppress baseline wander and noise. Furthermore, noise generated at the electrode–skin interface highlights the need for AFEs with high CMRR to ensure signal fidelity.

Future research is moving toward hybrid solutions that combine textile electrodes with nanostructured coatings, such as graphene or ZnO, to reduce impedance and improve biocompatibility. Other promising directions include integrating pressure sensors directly into the fabric to monitor electrode–skin contact in real time and developing new conductive polymers with enhanced long-term stability [76].

A comparative summary of electrode types and their key characteristics is reported in Table 1.

**TABLE 1.** Comparison of Reported ECG Standard Electrodes

| Feature | Wet | Dry | Textile-based |
|---|---|---|---|
| **Materials** | Ag/AgCl, conductive gel, hydrogel | Au, Ti, Pt, SS, PEDOT:PSS, CNT, graphene | Silver-plated nylon, copper-based fabric |
| **Advantages** | Excellent signal, low noise, clinical standard | Reusable, no gel, wearable-friendly | Soft, comfortable, washable, wearable-friendly |
| **Drawbacks** | Gel dries, skin irritation, single use | Motion artefacts, variable contact | Variable contact, humidity sensitive |
| **Typical use** | Clinical ECG, short-term tests | Long-term monitoring, wearables | Smart clothing, sports sensing |

### B. AMPLIFICATION STAGE

Amplification stage follows the sensor electrodes. Typically, it is an IA, which receives a differential signal and provides a voltage while rejecting common-mode interferences.

Traditionally, most biomedical AFEs rely on VM IAs. The classical three-op-amp IA architecture provides high input impedance, adjustable gain and CMRR, so making it suitable for clinical ECG and EEG monitoring [77]. Commercial devices such as the AD620 have been adopted in portable ECG systems, since they offer precision, simplicity, and cost-effectiveness [78]. VM IAs, however, face challenges when large electrode dc offsets are present, since these can saturate the amplifier input and limit the available dynamic range. Traditional dynamic compensation solutions such as ac coupling, autozeroing and chopper stabilization have been proposed to overcome this problem [79]. For instance, Yen et al. [27] proposed a micropower low-offset IA using an improved offset cancellation technique and achieving an input-referred rms noise voltage of 45 $\mu$V, and a current consumption of 61 $\mu$A.

In recent years, current-mode instrumentation amplifiers (CMIAs) have gained increasing attention in biomedical circuit design. Unlike VM architectures, CM approaches exploit current conveyors, flipped-voltage followers, or feedback loops to process biosignals in the current domain [80], [81]. These architectures offer several advantages, including intrinsically wider bandwidth, better linearity, and lower voltage headroom, which is beneficial in low-voltage CMOS technologies. For instance, Li et al. [82] demonstrated a CMIA with tunable gain controlled by a single resistor, achieving CMRR values above 120 dB at 1 Hz with power consumption around 20 $\mu$W in 0.35 $\mu$m CMOS [82]. More recently, Hoseini et al. [83] introduced a current-feedback IA with built-in offset cancellation loop for ECG/EEG, achieving 3.8 $\mu$Vrms input noise and 12.7 $\mu$W consumption in 180 nm CMOS.

### C. DRL ELECTRODE

To further suppress common-mode noise, especially from power line coupling, most modern ECG interfaces incorporate a DRL circuit. The DRL actively senses the common-mode voltage, present on the patient body, and injects a compensating signal via a third electrode (typically placed on the right leg). This negative feedback reduces common-mode voltages to the $\mu$V level, thereby substantially improving the SNR and the effective CMRR of the front-end amplifier.

In conventional three-electrode systems, the DRL sensor electrode provides a low-impedance return path that minimizes power line interference. Babusiak et al. [84] demonstrated that the addition of a DRL electrode in compact two- and three-electrode configurations significantly suppresses electromagnetic interference and prevents amplifier saturation, making it indispensable for portable and battery-powered ECG devices. When moving from wet to dry or capacitive electrodes, the role of the DRL becomes even more critical. Zompanti et al. [46] developed a portable ECG device based on dry capacitive electrodes and highlighted how the DRL loop was fundamental to reject common-mode voltages induced by body–power line coupling. Their results showed that the DRL enabled reliable multilead ECG recordings using a wearable shirt prototype, where without the DRL the signals would have been corrupted by motion artifacts and electromagnetic interference. Further improvements have been achieved by tailoring the DRL for specific electrode technologies. Guerrero et al. [85] proposed a high-gain DRL circuit specifically designed for dry electrodes, characterized by higher impedance and greater susceptibility to interference. Beyond incremental modifications, alternative approaches have also been explored. Guermandi et al. [45] introduced the Driving Right Leg (DgRL) circuit, an evolution of the classical DRL that leverages advanced feedback strategies to improve performance. Compared to simple grounding, the proposed DgRL achieves over 70 dB improvement in common-mode rejection and up to 30 dB more than traditional DRL implementations. This highlights the potential of rethinking the DRL principle for next-generation biomedical front-ends.

### *D. FILTERING STAGE*

The amplified signal is typically passed through a filtering block, designed to condition the waveform and remove undesired spectral components. ECG signals are weak, so block easily corrupted by various noise sources, including baseline wander, power line interference, electromyographic (EMG) activity and motion artifacts. The filtering stage is therefore fundamental to preserve the diagnostic content of the signal while attenuating spectral components that could obscure clinically relevant features [86]. Baseline wander, arising mainly from respiration and body movement, appears as a low-frequency drift (0.1–0.5 Hz) and can compromise interpretation of the ST segment. High-pass filters with cut-off frequencies between 0.05 Hz and 0.5 Hz are employed to mitigate this interference [86]. On the opposite end, EMG noise and environmental interference need a low-pass filtering, typically with cut-off frequencies in the 100–250 Hz range depending on the application [87]. A major challenge is power line interference at 50/60 Hz, which is usually addressed with dedicated notch filters or adaptive cancellation methods, as conventional bandpass filters alone are insufficient [64], [87]. Different strategies have been explored in the literature. Traditional approaches combine bandpass filters with notch filters for PLI suppression. However, these solutions can distort the phase of the ECG, a limitation particularly relevant for morphological analysis. For this reason, Waheed et al. [88] proposed a multistage filtering framework, where three cascaded additional stages for baseline wander removal, EMG suppression, and zero-phase filtering were adopted to achieve superior SNR improvement compared to single-stage filters.

More advanced approaches exploit adaptive filtering, which is well suited for nonstationary noise such as EMG. Tulyakova and Trofymchuk [89] introduced lightweight adaptive algorithms based on Savitzky–Golay filters and noise-dependent switching, showing that SNR improvements of up to 15 dB could be obtained while preserving waveform morphology.

At circuit design level, programmable AFEs integrate filtering directly within the analog chain. Han et al. [90] developed a programmable low-power AFE embedding a capacitor-coupled IA and a programmable low-pass filter with selectable cut-off frequencies (380 Hz, 451 Hz, and 1 kHz), implemented using current-steering techniques. This architecture enables flexible adaptation to different ECG monitoring contexts, from diagnostic-grade to wearable devices, while maintaining robustness against process variations.

## IV. VM AND CM APPROACHES

Previous considerations witness how the choice of circuit topology used in the input stage is decisive, as it directly impacts the noise performance, power consumption, and robustness of the system. Two main paradigms dominate the literature: VM and CM approaches. What distinguishes VM and CM approaches is not merely the type of active device employed, but also the way biosignals are represented and processed at circuit level. VM circuits rely on voltage differences across high-impedance inputs, while CM architectures translate signal information into currents that are subsequently manipulated by dedicated building blocks such as CCII or VCII.

### *A. VM BIOSIGNAL AMPLIFIERS*

VM architectures have historically represented the mainstream solution for ECG front ends due to their intrinsically high input impedance, well-established compatibility with legacy clinical instrumentation and the maturity of CMOS implementations. Their evolution in the last two decades has focused on balancing ultralow power consumption, reduced noise, and robustness to large electrode offsets, while maintaining CMRR compatible with clinical standards.

Early CMOS contributions already recognized the importance of a fully differential structure combined with on-chip dc suppression. Spinelli et al. [47] demonstrated that a fully differential VM amplifier with dc suppression could mitigate electrode offset without resorting to large discrete components, thereby enabling faster recovery and high CMRR at reduced supply voltages. This trend toward dc-coupled or pseudoresistive suppression has been consolidated by later works. Zhao et al. [48], for example, presented a dc-coupled fully differential difference amplifier (FDDA) achieving an NEF of 2.55 with a CMRR of 76 dB, highlighting the potential of VM designs. The CCIA has been a cornerstone of VM front-end design because it allows the implementation of sub-Hz high-pass cut-off without physically large resistors. Ng et al. [33] introduced a single-stage CCIA with complementary transimpedance boosting, demonstrating that pseudoresistors can be effectively combined with circuit-level innovations to optimize both noise and silicon area. This approach has been further extended in recent designs: Laababid et al. [91] presented a low-power CMOS ECG amplifier where pseudoresistive elements were exploited to realize ultralow high-pass cut-off frequencies, while also highlighting the practical limitations of pseudoresistor implementations in terms of linearity and stability under time-varying electrode offsets, while Song et al. [92] proposed a dc-servo loop realized with series-switch pseudoresistors to achieve more stable and linear high-pass filtering. These works illustrate that although pseudoresistive ac coupling remains central in VM ECG amplifiers, its limitations are increasingly being addressed through refined circuit strategies.

Ultralow supply operation has also been a persistent theme. Bai et al. [37] demonstrated that a 0.5 V chopper-stabilized CMOS IA could maintain low IRN while consuming sub-$\mu$W power, proving the feasibility of VM architectures in near-threshold conditions. Similarly, Dong et al. [20] reported a programmable-gain VM amplifier operating at 0.6 V supply that achieves an exceptionally high CMRR of 119 dB while maintaining a low NEF, setting a benchmark for ultralow voltage biopotential front-ends. More recently, Fan et al. [35] showed that even in deeply scaled 40-nm CMOS, a three-stage chopper-stabilized DDA operating at 0.5 V is able to give an 89 dB open-loop gain with a power consumption of only 0.74 $\mu$W, reinforcing the relevance of VM techniques at scaled supply voltages.

In addition to noise and offset suppression, contemporary research has emphasized extreme power efficiency. Sawigun and Thanapitak [93] presented a compact sub-$\mu$W CMOS ECG amplifier based on a fully differential DDA, fabricated in 0.35 $\mu$m technology, which operates consuming only 336 nA at a 2 V supply while achieving an input impedance lower than 10 GΩ, a NEF of 2.02, and a CMRR of 83 dB. This result confirmed that VM amplifiers can sustain competitive energy efficiency metrics while preserving input impedance requirements, a key factor for dry and textile electrodes.

Beyond single-channel demonstrations, system-level integrations have validated the robustness of VM approaches. Deepu et al. [94] described an ECG-on-chip that combines a VM IA with bandpass filtering and an integrated SAR ADC, achieving a 9.6 $\mu$W dissipation while maintaining diagnostic-grade signal fidelity. Such works illustrate that VM front-ends are not limited to stand-alone amplifiers but can be effectively embedded in complete system-on-chip solutions for wearable or portable ECG monitoring.

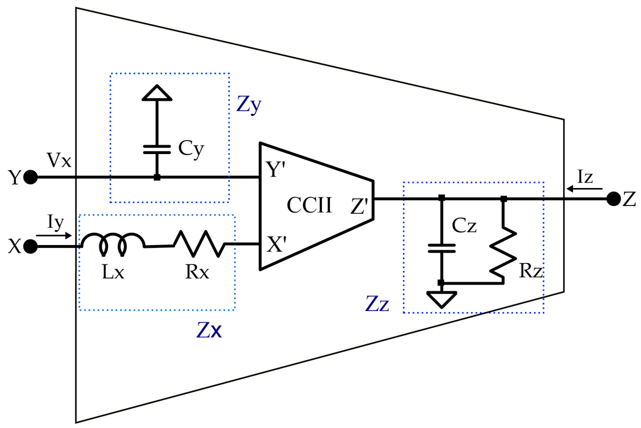


**FIGURE 3.** Second-generation current conveyor with parasitic components.

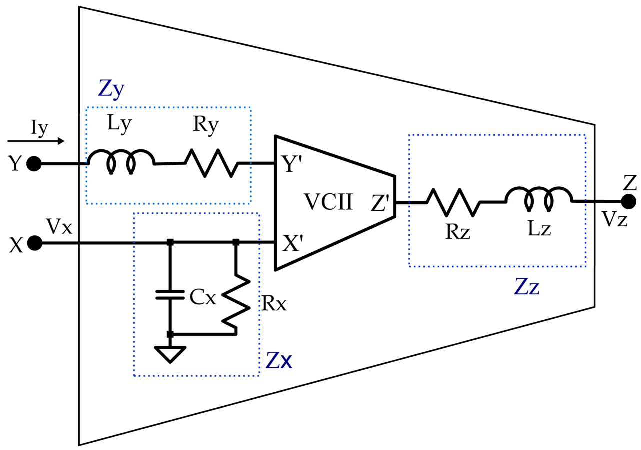


**FIGURE 4.** VCII with parasitic components.

### B. CM BIOSIGNAL AMPLIFIERS

Traditional reported VM methods in designing read-out circuits suffer from several drawbacks, such as complexity, large chip area, and high power consumption. The CM techniques have shown in some cases the operational amplifier is not the best active block to be used because some of its performances are worse than those offered by the most common CM counterparts [95]. The main CM analog blocks are the CCII and its dual, the VCII. These blocks have been successfully applied to the realization of a variety of analog signal processing functions such as ECG and biosignal analog processing [96], [97], [98], [99]. CCII is the most well-known CM analog active block; An ideal CCII is a three-terminal device (X, Y, Z) that behaves as a voltage buffer from Y to X and as a current buffer from X to Z. Fig. 3 shows the symbol of CCII with parasitic components at its nodes. In this configuration: Y is a high-impedance (ideally infinite) voltage input terminal, X is a low-impedance (ideally zero) current input/voltage output terminal, Z is a high-impedance (ideally infinite) current output terminal. The real voltage-current relationships of a CCII are defined as follows:

$$\begin{bmatrix} I_Y \\ V_x \\ I_z \end{bmatrix} = \begin{bmatrix} sC_y & 0 & 0 \\ \alpha & (r_x + sL_x)//\frac{1}{sC_x} & 0 \\ 0 & \pm\beta & \frac{1}{r_z//\frac{1}{sC_z}} \end{bmatrix} \begin{bmatrix} V_y \\ I_x \\ V_z \end{bmatrix} \quad (1)$$

being $C_y$, $r_x$, $L_x$, $C_x$, $r_z$, and $C_z$ the parasitic capacitance at Y terminal, the parasitic resistance, inductance, and capacitance associated with X terminal and the parasitic resistance and capacitance at Z terminal, respectively.

In the following we analyze some of the several works in the literature that have already demonstrated the suitability of CM approach and CCII-based front-ends for electrocardiography, electroencephalography (EEG) and other low-frequency biopotential recordings.

Wang et al. [55] presented a CM capacitively coupled chopper instrumentation amplifier specifically optimized for biopotential acquisition with both resistive and capacitive electrodes, achieving a power consumption of 4 $\mu$W per channel and an input referred noise of 160 nV/$\sqrt{}$ Hz. The proposed design demonstrated that combining capacitive coupling with CM operation alleviates the effect of electrode offset while maintaining low IRN. The chopper stabilization further reduced flicker noise, highlighting the potential of CM architectures to compete with and even outperform VM chopper amplifiers under comparable supply and noise constraints.

The flexibility of CCII-based designs was shown earlier by Stornelli and Ferri [56], who proposed a single-CCII bootstrap circuit for ECG and EEG applications. Their approach enabled high input impedance while maintaining low-voltage and low-power operation, thus extending the usability of CCII to systems requiring electrode buffering and impedance boosting.

This bootstrap configuration demonstrated that even a minimal CCII core could effectively mitigate the impedance mismatch issues that typically complicate the integration of dry or high-impedance electrodes.

More recently, Pantuprecharat et al. [65] presented a CCII-based ECG amplifier with integrated dc suppression, presenting an input referred noise of 1.62 $\mu$Vrms.

By leveraging the inherent CM properties of the CCII and embedding a dc cancellation loop, they showed that baseline wander and electrode offset could be effectively rejected with a CM approach without compromising signal quality. This recent work underscores the continued relevance of CCII in addressing longstanding challenges of ECG front-ends, particularly when electrodes with variable impedance are employed in dynamic or wearable conditions.

Another CM active block is the VCII, that is nothing more than the dual of CCII, and for this reason it enjoys almost all the properties of CCII but also provides an easy-to-read voltage as an output signal. As for the CCII, the information in a VCII is carried both by currents and voltages, but VCII is equipped with both an input and an output voltage port. This represents one of the main advantages with respect to CCII, because voltage measurements are generally easier. In Fig. 4 the symbol of VCII with parasitic components at its nodes is shown. An ideal VCII works as a current buffer from Y to X and as a voltage buffer from X to Z. Y is a low-impedance (ideally zero) current input terminal, X is a high-impedance (ideally infinite) voltage input and current output terminal, Z is a low-impedance (ideally zero) voltage output terminal. The nonideal voltage-current relationships of a VCII are defined by the following matrix:

$$\begin{bmatrix} I_x \\ V_y \\ V_z \end{bmatrix} = \begin{bmatrix} \frac{1}{r_x} + sC_x & \pm\beta & 0 \\ 0 & r_y + sL_y & 0 \\ \alpha & 0 & r_z + sL_z \end{bmatrix} \begin{bmatrix} V_x \\ I_y \\ I_z \end{bmatrix} \quad (2)$$

where $r_x$ and $C_x$ represent the parasitic resistance and capacitance at X terminal, $r_y$ and $L_y$ are the parasitic resistance and inductance at Y terminal, and finally, $r_z$ and $L_z$ are the parasitic resistance and inductance associated with Z terminal, respectively.

Although the VCII is a relatively recent circuit element, its structural properties make it attractive for AFEs in biomedical applications, where both current and voltage signals must be handled efficiently and the final output is often required in VM for compatibility with subsequent stages. Its capability of operating at low supply voltages and of providing a low-impedance voltage output has been exploited in different biosignal processing circuits.

In [52], a general-purpose biomedical amplifier by combining a differential flipped voltage follower (DFVF) with a VCII was demonstrated. The resulting topology exhibited great input referred noise (5.4 $\mu$Vrms) compatible with ECG and EEG acquisition, very low power consumption (20 $\mu$W), and reliable operation at supply voltages as low as ±0.6 V in 0.15-$\mu$m CMOS technology. A more direct validation in ECG acquisition was later provided in [64], in this work a complete VCII-based readout circuit for wearable cardiac monitoring was designed. Their results showed that the low output impedance at the Z terminal simplifies the interfacing with subsequent processing blocks while preserving diagnostic quality of the acquired ECG waveforms, highlighting a CMRR of 85 dB.

The versatility of VCII in biosignal processing has also been highlighted by Kumngern et al. [100], who implemented a low-pass filter for biosensor applications operating at a 0.5 V supply. The proposed design confirms that VCII-based topologies can sustain linear and low-power analog conditioning functions even under strict energy constraints with a power consumption of 2.73 $\mu$W and the total harmonic distortion is below 1% .

To conclude this section, a comparative overview of several state-of-the-art amplifiers and analog interface circuits developed for ECG signal acquisition is presented in Table 2. The purpose of this summary is to highlight the main performance trends emerging from recent literature, emphasizing the differences between VM and CM implementations. The comparison includes main features and specification such as technology node, supply voltage, power consumption, IRN, and CMRR. This analysis provides a clear picture of the

**TABLE 2.** Summary of Some Reported ECG AFE and Amplifiers With the Related Objectives, Main Features, and Specifications

| Reference | Objective | Main Features | Specifications |
|---|---|---|---|
| [19] | To design and analyze a high-gain, low-noise, low-power AFE for ECG acquisition using the $g_m/I_D$ methodology. | AFE includes an IA and cascaded filters (fifth-order LPF, second-order HPF, Notch filter) all based on two-stage Miller-compensated OTAs. | Process: 45 nm TSMC CMOS. Supply: ±0.6 V. Power: 10.88 $\mu$W. Gain: 58 dB (tunable). IRN: 33.6 $\mu$Vrms (integrated). Approach: Voltage mode |
| [20] | To propose a Programmable Gain Amplifier with common-mode control for robust ECG recording. | A novel common-mode voltage control module employs a VCO-based comparator and improved PFD to enhance the CMRR. Gain adjustment is achieved by modifying input capacitance, and the unity-gain buffer utilizes Quasi-Floating Gate technology for low quiescent power | Process: 65 nm CMOS. Supply: 0.6 V. Power: 9.18 $\mu$W. CMRR: 119.2 dB. IRN: 6.6 $\mu$Vrms. Approach: Voltage mode. |
| [30] | To present a three Op-Amp IA with active dc suppression for biopotential measurements. | DC suppression is achieved via a controlled floating source at the input stage, implemented using an optical-isolated device (optocoupler). The architecture preserves the high input impedance and high CMRR characteristics of the classic three op-amps IA. | Gain: 80 dB. CMRR: 105 dB. Approach: ac-coupled, 3-Op-Amp IA (Voltage mode). |
| [32] | To present a digitally programmable CMOS AFE IC for portable EEG/ECG monitoring applications operating at reduced supply voltage. | AFE is constituted by a new rail-to-rail IA composed of parallel chopper-stabilized DDAs (pMOS/nMOS stages) to achieve a wide input common-mode range. Chopper stabilization is used to remove flicker noise and offset | Process: 0.5 $\mu$m CMOS. Supply: 1.5 V. Current: 485 $\mu$A. CMRR: ≥115 dB (0–1 kHz). IRN: 0.86 $\mu$Vrms (0.3–150 Hz). Approach: Voltage mode. |
| [46] | To characterize and implement an electronic interface for a portable ECG device using dry capacitive electrodes and a DRL circuit. | The circuit architecture utilizes dry capacitive electrodes (contactless ECG device) for long-term monitoring, aiming for embedding in clothes. It also employs a DRL circuit to reject Common-Mode Interference and ensures high input impedance via a double buffering stage | Measured gain: 60 dB. Component: AD8232. Approach: Voltage mode |
| [47] | To describe a novel fully differential biopotential amplifier with dc suppression suitable for low supply voltage systems. | The solution is based on a fully differential active dc-suppression circuit that avoids matched passive components, offering high CMRR and fast recovery from transients. Uses standard low-power op amps. | Supply: 5 V. CMRR: 102 dB. DC Input Range: ≤500 mV. Approach: Voltage mode. |
| [48] | To design an ultralow power dc-coupled AFE for wearable biomedical sensors with high input impedance and high CMRR. | The interface is formed by an FDDA topology. It introduces a parasitic capacitor reuse technique in large input transistors to improve noise and area efficiency. It employs an on-body dc bias scheme to manage input dc offset. | Process: 0.35 $\mu$m CMOS. Supply: 1.8 V. Power: 0.9 $\mu$W (499 nA). CMRR: 76 dB. IRN: 1.02 $\mu$Vrms. Approach: Voltage Mode. |
| [52] | To propose a low complexity mixed operation mode amplifier for high integration level, especially for biomedical systems. | Combines a differential flipped voltage follower (DFVF) and a VCII in a mixed voltage/current mode architecture. It features high input impedance due to bootstrap operation, enabling ac coupling with small capacitors | Process: 150 nm CMOS. Supply: ±0.6 V. Power: 20 $\mu$W. Gain: 0–33 dB. IRN: 5.4 $\mu$Vrms (0.1 Hz–10 kHz). Approach: Voltage + Current Mode (DFVF/VCII). |
| [55] | To present a high-density, low-noise AFE for capacitively coupled biosignal recording. | Proposes a Current-Mode Capacitively Coupled Chopper IA, where choppers are placed after coupling capacitors but outside the feedback loop. This architecture avoids 1/f IRN typically seen in ac-coupled chopper systems | Process: 180 nm SOI CMOS. Power: 4.8 $\mu$W/channel. IRN Density: 160 nV/√Hz. Approach: Current mode |
| [56] | To propose a novel low voltage, low power single-CCII bootstrap circuit specifically for ECG/EEG input stages. | This simplified integrable bootstrap solution utilizes a single CCII, reducing the component count compared to previous two-CCII versions. The design uses cascoded output transistors to achieve high Z node parasitic resistance | Supply: 1.5 V. Power: 28 $\mu$W. IRN Density: 18 $\mu$V/√Hz (@1 kHz). Input Impedance: 62 MΩ. Approach: Current mode |

**TABLE 2. Continued**

| | | | |
|---|---|---|---|
| [63] | To present a novel CMIA using the OFCC as basic building block. | The read-out circuit employs two OFCCs to achieve high differential gain and a bandwidth that is independent of gain. It maintains a high CMRR and superior accuracy without needing matched resistors | Gain: 40 dB (tunable). CMRR: 76 dB. Slew rate: 395 V/$\mu$s. Approach: Current mode. |
| [64] | To present the design of a read-out circuit for ECG monitoring using VCIIs as active blocks. | The interface uses VCIIs, incorporating a DRL electrode for common-mode rejection and features tunable differential gain. | Process: 0.35 $\mu$m CMOS. Supply: ±3.3 V. Gain: 70 dB. CMRR: 85 dB. (tunable) IRN: 1.372 $\mu$V. Approach: Current mode. |
| [65] | To design a compact CCII-based ECG amplifier that incorporates dc suppression operation. | The design uses two CCIIs and passive components (R and C) forming the amplifier circuit. It utilizes the current-controlled current source CCII for compact implementation | Components: AD844. Gain: 46 dB. CMRR: 57.76. IRN: 1.62 $\mu$Vrms. Approach: Current mode. |
| [66] | To develop a low-noise, low-power amplifier based on a classical amplifier structure for Wearable ECG sensors | The solution utilizes capacitive feedback and MOS-bipolar pseudoresistors to reject dc offset and achieve required low-frequency bandwidth. The structure is simpler and employs less transistors than classic Op-Amps | Process: 180 nm CMOS. Supply: 1 V. Power: 1.44 $\mu$W. Gain: 38.43 dB. CMRR: 87.72 dB NEF: 18.07. Approach: Voltage Mode. |
| [69] | To design a low noise and low power capacitive-feedback amplifier for ECG recordings utilizing a current-reused OTA | This solution employs an OTA with an inverter-based differential input stage and a Class-AB output stage for high efficiency. Current reuse is achieved by embedding the Class-AB control unit into the input stage, thus reducing power consumption | Process: 0.35 m CMOS. Supply: 2 V. Power: 0.32 $\mu$W (160 nA). CMRR: 65 dB. IRN: 2.05 $\mu$Vrms. NEF: 2.26. Approach: Voltage Mode. |
| [82] | To describe the design of a CMIA for portable biosignal acquisition systems | The interface is formed by a CMIA topology based on the Op Amp power supply current sensing technique. It offers continuously adjustable voltage gain, independent of GBW | Process: 0.35 $\mu$m CMOS. Supply: 3 V. Power: 20.22 $\mu$W. CMRR: >120 dB (<1 Hz). IRN Density: 170 nV/√Hz. Approach: Current mode |
| [83] | To describe a compact and low-power Current Feedback Instrumentation Amplifier (CFIA) that eliminates Differential Electrode Offset (DEO) and input offset | This solution implements a local feedback loop in the CFIA first stage to eliminate dc offset, allowing offset cancellation with low bias current. | Process: 0.18 $\mu$m CMOS. Supply: 1.8 V. Power: 12.7 $\mu$W. CMRR: 120 dB (no DEO) / 95 dB (250 mV DEO). IRN: 3.8 $\mu$Vrms. Approach: Current-Mode |
| [90] | To propose a programmable low power and low noise AFE for ECG signal acquisition | The interface is formed by a CCIA featuring independent biasing technique and ac coupling for stable static working point and noise performance. Utilizes current steering technology for an area-efficient programmable low pass filter | Process: 0.13 $\mu$m CMOS. Supply: 1.2 V. Power: 48 $\mu$W. CMRR: 93 dB. IRN: 0.6 $\mu$V (integrated). Approach: Voltage mode. |
| [91] | To design and simulate an AFE IC for recording ECG signals in smart wearable devices. | The interface uses a current-reused OTA with an inverter-based differential input stage and Class-AB output stage for low noise and high efficiency. It includes a DRL circuit and filtering blocks (Notch, HPF) to reduce PLI and other contaminants | Process: 180 nm CMOS. Supply: 1.8 V. Power: 1.08 $\mu$W. Gain: 50.75 dB. CMRR: 102 dB. IRN: 1.47 $\mu$Vrms (0.1–1 kHz). NEF: 2.74. Approach: Voltage Mode. |
| [93] | To propose a compact DDA-based CMOS IA suitable for micropower ECG monitoring | The solution features an energy-efficient current-sharing DDA realized with only eight transistors. It is dc-coupled via gate terminals for high input impedance and is self-stabilized at dc via pseudo resistors, eliminating the need for CMFB | Process: 0.35 $\mu$m CMOS. Supply: 2 V. Power: 0.672 $\mu$W (336 nA). CMRR: 83.24 dB. IRN: 1.54 $\mu$Vrms (0.1–300 Hz). NEF: 2.02. Approach: Voltage Mode. |

design trade-offs adopted by various authors and serves as a reference point for identifying the most suitable approaches for low-power and high-fidelity ECG front-end design.

The comparative data in Table 2 clearly show that both VM and CM architectures can achieve competitive performance in terms of CMRR, IRN, and power consumption. However, VM architectures remain the standard in commercial ECG systems due to their higher input impedance and well-established CMOS implementations. They are ideal for clinical-grade precision and compatibility with wet electrodes.

Conversely, CM architectures based on CCII, VCII, or operational floating current conveyors (OFCC) exhibit superior bandwidth stability, simpler gain programmability, and higher slew-rate, making them attractive for wearable and portable systems where fast signal processing is needed. Among these, VCII-based designs appear particularly promising because they combine CM processing with voltage output compatibility, simplifying integration with digital or mixed-signal subsystems. As technology scales and energy efficiency becomes paramount, hybrid voltage–current architectures are expected to dominate future ECG front-end design, filling the gap between performance and integration capability.

## V. CONCLUSION

This review has outlined the evolution of AFE circuits and interface technologies for ECG signal acquisition, analyzing their operating principles and performance trade-offs. Future progress in this field is expected to emerge from the convergence of three main directions: Advanced CMOS circuit design, mixed-signal system integration, and innovation in electrode technologies.

First, advanced CMOS processes will increasingly play a main role in the development of ultralow-power and high-integration AFEs. Dynamically reconfigurable AFEs are beginning to gain attention, capable of adapting their power and noise characteristics in real time according to signal conditions, [101]. The scaling toward 65 nm and 28 nm CMOS nodes will further enable the adoption of near-threshold and digitally assisted techniques, allowing analog designers to optimize the noise–power trade-off with unprecedented flexibility.

Second, system-level codesign between the AFE and subsequent signal processing stages is gaining growing attention. Compressed-sensing-based AFEs have been proposed to reduce the data rate at the acquisition stage while preserving diagnostic information, thus minimizing the power required for data transmission and storage [102]. This paradigm shift toward analog–digital optimization will be decisive for next-generation wearable and implantable ECG systems, where the boundaries between sensing, analog conditioning, and digital feature extraction become increasingly blurred.

Third, significant advances are expected from the electrode–circuit interface, driven by the emergence of nanostructured and biocompatible materials. Flexible electrodes based on carbon nanotubes, graphene, or silver nanowires integrated into polymeric or textile substrates have shown excellent conformability and stability for long-term ECG monitoring [103]. Hybrid electrodes combining conductive polymers with nanomaterials are now being developed to reduce electrode–skin impedance and motion artifacts, ensuring higher comfort and durability in wearable applications [104]. Furthermore, recent “gentle-to-skin” printed electrodes have demonstrated the possibility of directly integrating sensing and electronic layers on soft, stretchable materials, paving the way toward fully integrated epidermal ECG sensors [105].

Beyond the directions already discussed, future research in ECG front-end architectures is expected to evolve toward the integration of artificial intelligence (AI)-assisted signal conditioning and machine learning-based noise reduction directly embedded in mixed-signal circuits. The combination of on-chip learning and adaptive filtering can enable personalized ECG acquisition systems capable of self-calibrating to different electrode types and skin conditions.

Another emerging trend concerns flexible hybrid electronics, merging organic and inorganic materials to realize conformable front-ends that can operate at ultralow voltages ($<0.5$ V). These systems could be directly printed or deposited onto flexible substrates or textiles, allowing seamless integration into wearable garments.

Moreover, energy harvesting from body heat or motion is being explored to create autonomous ECG monitoring nodes, paving the way toward *battery-free* biosignal acquisition.

Finally, the adoption of neuromorphic analog architectures could represent a paradigm shift for on-sensor processing, enabling real-time anomaly detection with minimal power consumption. These trends collectively define a roadmap toward next-generation *smart AFEs* that merge low-power electronics, intelligent adaptation, and biocompatible integration.

In conclusion, the future of ECG AFEs lies in the synergy between low-power CMOS design, system-level optimization, and biocompatible material innovation. The integration of adaptive AFEs, codesigned analog–digital architectures, and nanostructured, flexible electrodes will enable a new generation of wearable and implantable ECG monitoring systems—capable of continuous, unobtrusive, and reliable operation in real-life conditions.

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